Vincent Zheng

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Education

Stony Brook University — Stony Brook, NY
M.S. in Computer Science - Accelerated B.S./M.S. Program — *Expected May 2028*
B.S. in Computer Science | GPA: 3.76 / 4.00 — *Expected May 2027*

Experience

Software Engineer Intern — Raleigh, NC
First Citizens Bank — *May 2026 - July 2026*

- Engineered a Node.js microservice to automate AEM site onboarding, generating brand configurations and demo content packages on demand and **cutting setup time from days to minutes.**
- Built multi-threaded structured test-execution logging into Splunk for a Java-based test automation platform serving **7 QA teams**, centralizing failure metrics and eliminating manual log inspection.
- Refactored **10+ AEM components** with Java Sling Models, decoupling backend logic from HTL templates.
- Designed JUnit test suites with **95% code coverage** across component models, validating backend logic.

Security Research Assistant — Stony Brook, NY
Ethos Security & Privacy Labs — *Jan 2026 - Present*

- Engineered an SSH honeypot backend combining deterministic shell emulation with LLM-based fallback generation, increasing attacker engagement from **2.96 to 5.41 commands per session.**
- Developed a SQLite persistence layer caching LLM-generated file contents in the virtual filesystem and mapping sessions by IP, enabling stateful attacker interactions while **reducing API calls by 30%.**
- Implemented CI/CD pipelines with GitHub Actions, automating testing and container releases.

Software Engineer Intern — Stony Brook, NY
Stony Brook University Web Services — *Sep 2025 – May 2026*

- Built **15+ responsive reusable components** using HTML/CSS/JavaScript in Modern Campus CMS and set up **20+ new department sites** by configuring templates, files, navigation, and settings for content owners.
- Streamlined CMS operations for **8 engineers** through a suite of custom automation gadgets, eliminating repetitive workflows and saving an estimated **20+ engineering hours monthly**.
- Authored **5+ technical guides and video tutorials** for future developers, reducing onboarding ramp-up time.

Publications

Hybrid Cowrie: A Stateful SSH Honeypot with Shell Execution and LLM Fallback

- **V. Zheng**, R. Hung, A. Pitumpe, A. Rahmati. Manuscript in preparation.

Multi-Agent Proximal Policy Optimization for Network-Aware Drone-based Crop Image Analysis

- A. Hellman, B. Wagle, S. Peppers, **V. Zheng**, et al. To appear in *IJCNN 2026*.

Projects

Online Book Store — *Spring Boot, Spring MVC, Thymeleaf, Hibernate, MySQL, Stripe API*

- Developed an online bookstore using Spring Boot, Spring MVC, Thymeleaf, Hibernate, and MySQL, allowing sellers to list and manage inventory and buyers to browse, search, and purchase books.
- Implemented Spring Security for authentication and access control, and integrated Stripe for secure checkout.

Social Media Platform — *React.js, Node.js, Express, MongoDB*

- Built a Reddit clone with nested comments, community/post creation, and search across posts and comments.
- Designed MongoDB schemas and REST API routes with Axios integration for frontend/backend communication.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C, SQL, HTML, CSS
Tools: AWS, Git, Docker, Bash, Jest, Pytest
Frameworks: Spring Boot, MERN stack (MongoDB, Express.js, React.js, Node.js), FastAPI
Certifications: AWS Solutions Architect - Associate, AWS Cloud Practitioner